

A scalar operator-valued measure with no separable dilation

Full negative answer to Problem B of arXiv:1110.5833

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Status. Candidate full solution, likely valid, pending human review. The construction answers both parts of the source problem negatively.

1 Source question

Han, Larson, Liu, and Liu develop Banach-space dilations of operator-valued measures in [1]. Their Problem B appears in Section 2.4, printed page 37 (PDF page 41), after they prove separability for the discrete framing case and before Proposition 2.38.

The above example shows that in the case that the operator-valued probability measure system is induced by a framing $\{x_i, y_i\}_{i \in \mathbb{N}}$ of X , then the minimal α -dilation space is exactly the one constructed in **CHL** and hence it is always separable. However, it is not clear whether this is true in general. So we ask:

Problem B. Assume that $(\Omega, \Sigma, E, B(X))$ is an operator-valued measure system such that X is a separable Banach space. Is the dilation space α equipped with the minimal dilation norm always separable? If not, does there exist a dilation space which is separable?

Figure 1: Problem B in the source paper, rendered from `source_paper.pdf`, PDF page 41.

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We answer both questions negatively, already for the one-dimensional Banach space $X = \mathbb{C}$ and a positive scalar multiple of the identity.

2 Definitions

For an operator-valued measure $E : \Sigma \rightarrow B(X)$, the source's algebraic elementary dilation space is spanned by the set functions

$$E_{B,x}(A) = E(A \cap B)x \quad (A, B \in \Sigma, x \in X).$$

Its minimal dilation norm is

$$\left\| \sum_{k=1}^n c_k E_{B_k, x_k} \right\|_{\alpha} = \sup_{A \in \Sigma} \left\| \sum_{k=1}^n c_k E(A \cap B_k) x_k \right\|_X. \quad (1)$$

The minimal dilation space is the completion in this norm.

A dilation of E consists of a Banach space Z , bounded maps $T : X \rightarrow Z$ and $S : Z \rightarrow X$, and a countably additive projection-valued measure $F : \Sigma \rightarrow B(Z)$ such that

$$E(B) = SF(B)T \quad (B \in \Sigma). \quad (2)$$

Projection-valued measures in the source are spectral: $F(A)F(B) = F(A \cap B)$.

3 Proof intuition

The obstruction is not the dimension of X but the density of the underlying measure algebra. An uncountable Bernoulli product has uncountably many coordinate events separated by a fixed amount of measure. In the minimal model, their elementary vectors inherit this separation directly from (1).

Changing the dilation cannot compress the obstruction into a separable space. Indeed, every coordinate event produces a vector $F(B_i)T1$. If two such vectors were close, applying the spectral projection of the one-sided difference $B_i \setminus B_j$ would isolate a vector whose scalar coefficient is exactly $\mu(B_i \setminus B_j) = 1/4$. Uniform boundedness of the spectral projections turns this into a fixed norm separation.

4 Full negative solution

Theorem 1. *There exist a measurable space (Ω, Σ) , a separable Banach space X , and an operator-valued probability measure $E : \Sigma \rightarrow B(X)$ such that*

1. *the minimal dilation space of E is nonseparable; and*
2. *every projection-valued dilation space of E is nonseparable.*

Consequently both questions in Problem B of [1] have negative answers.

Proof. Let I be an uncountable set and take the fair Bernoulli product probability space

$$(\Omega, \Sigma, \mu) = (\{0, 1\}^I, \text{ product } \sigma\text{-algebra}, (\tfrac{1}{2}\delta_0 + \tfrac{1}{2}\delta_1)^I).$$

Let $X = \mathbb{C}$ and define

$$E(B) = \mu(B)I_{\mathbb{C}} \quad (B \in \Sigma).$$

This is a strongly countably additive $B(\mathbb{C})$ -valued probability measure, and X is separable.

For $i \in I$, put

$$B_i = \{\omega \in \Omega : \omega_i = 1\}, \quad u_i = E_{B_i, 1}.$$

If $i \neq j$, independence of the two coordinates gives

$$\mu(B_i \setminus B_j) = \frac{1}{4}. \quad (3)$$

Using $A = B_i \setminus B_j$ in (1), we obtain

$$\begin{aligned}\|u_i - u_j\|_\alpha &\geq |E(A \cap B_i)1 - E(A \cap B_j)1| \\ &= \mu(B_i \setminus B_j) = \frac{1}{4}.\end{aligned}$$

Thus the algebraic minimal dilation space already contains the uncountable 1/4-separated family $(u_i)_{i \in I}$. Its completion is nonseparable.

It remains to show that no other dilation can be separable. Let (Z, F, S, T) be an arbitrary dilation as in (2). Since $X = \mathbb{C}$, write $z = T1 \in Z$ and regard S as a functional in Z^* . Then

$$S(F(B)z) = \mu(B) \quad (B \in \Sigma). \quad (4)$$

The family of projections is uniformly bounded:

$$M := \sup_{B \in \Sigma} \|F(B)\| < \infty. \quad (5)$$

Indeed, for each $y \in Z$, the map $B \mapsto F(B)y$ is a countably additive Z -valued measure, hence has bounded range; the uniform boundedness principle then gives (5).

Set $w_i = F(B_i)z$. Fix distinct i, j and let $D = B_i \setminus B_j$. Spectrality gives

$$F(D)(w_i - w_j) = F(D \cap B_i)z - F(D \cap B_j)z = F(D)z.$$

Together with (4), (3), and (5), this yields

$$\|w_i - w_j\| \geq \frac{\|F(D)z\|}{M} \geq \frac{|S(F(D)z)|}{M\|S\|} = \frac{1}{4M\|S\|}.$$

Here $S \neq 0$ because $S(F(\Omega)z) = \mu(\Omega) = 1$. Therefore $(w_i)_{i \in I}$ is an uncountable uniformly separated family in Z , so Z cannot be separable. This proves both assertions. $\square$

5 Verification and scope

The proof has no numerical or unproved dependency. Its checkable points are:

1. $X = \mathbb{C}$ satisfies the separability hypothesis and $E(\Omega) = I_{\mathbb{C}}$;
2. distinct fair Bernoulli coordinates satisfy $\mu(B_i \setminus B_j) = 1/4$;
3. the displayed minimal norm is exactly the source's definition;
4. strong countable additivity of F implies pointwise boundedness of $\{F(B) : B \in \Sigma\}$, and Banach–Steinhaus makes it uniformly bounded;
5. spectrality isolates $F(B_i \setminus B_j)z$ from $w_i - w_j$.

The construction uses a non-countably-generated measure algebra. It therefore does not contradict Proposition 2.38 of the source, which obtains separability from a countable Borel basis together with total strong countable additivity. Rather, the example shows that some restriction of this kind is essential if one wants a general separability theorem.

6 Bounded novelty check

On 9 August 2026 we searched the run’s registry, solution, attempt, and proof-gap indexes; the local arXiv corpus; exact phrases from Problem B; and web searches combining “minimal dilation space”, “operator-valued measure”, and “separable dilation”. The search found the source and the authors’ later exposition arXiv:1411.4559, but no claimed solution of Problem B. The later exposition repeats several other memoir problems while omitting this one. This is evidence, not a proof of novelty; the result should receive expert literature review.

7 Human-review recommendation

Review as a candidate full negative answer. The main verifier focus should be the intended generality of Problem B (arbitrary measurable spaces) and the standard bounded-range/uniform-boundedness step for F . If both are accepted, the argument closes the entire stated problem rather than merely a subcase.

References

- [1] D. Han, D. R. Larson, B. Liu, and R. Liu, *Operator-Valued Measures, Dilations, and the Theory of Frames*, arXiv:1110.5833 (2011); *Memoirs of the American Mathematical Society* **229**, no. 1075.